

DURGAPRASAD V

Project Title

Customer Churn Prediction Using Machine Learning

Objective

Predict whether a customer will leave a company (Churn = Yes/No) using supervised machine learning.

Dataset

Use the popular **Telco Customer Churn Dataset**

[IBM Telco Customer Churn Dataset](#)

Algorithms to Compare

1. Logistic Regression
 2. Random Forest
-

Steps to Implement

1. Import Libraries

```
None
import pandas as pd
import numpy as np

from sklearn.model_selection import train_test_split
from sklearn.model_selection import cross_val_score
```

```
from sklearn.preprocessing import LabelEncoder
from sklearn.metrics import accuracy_score
from sklearn.metrics import precision_score
from sklearn.metrics import recall_score
from sklearn.metrics import f1_score
from sklearn.metrics import roc_auc_score
from sklearn.metrics import confusion_matrix

from sklearn.linear_model import LogisticRegression
from sklearn.ensemble import RandomForestClassifier
```

2. Load Dataset

```
None
df = pd.read_csv("WA_Fn-UseC_-Telco-Customer-Churn.csv")
df.head()
```

3. Data Preprocessing

Check Missing Values

```
None
df.isnull().sum()
```

Convert Categorical Columns

```
None
le = LabelEncoder()

for col in df.columns:
    if df[col].dtype == 'object':
        df[col] = le.fit_transform(df[col].astype(str))
```

4. Split Features and Target

None

```
X = df.drop("Churn", axis=1)
y = df["Churn"]
```

5. Train-Test Split

None

```
X_train, X_test, y_train, y_test = train_test_split(
    X, y,
    test_size=0.2,
    random_state=42
)
```

6. Logistic Regression

None

```
lr = LogisticRegression(max_iter=1000)

lr.fit(X_train, y_train)

y_pred_lr = lr.predict(X_test)
y_prob_lr = lr.predict_proba(X_test)[:,-1]
```

Metrics:

None

```
print("Accuracy:", accuracy_score(y_test, y_pred_lr))
print("Precision:", precision_score(y_test, y_pred_lr))
print("Recall:", recall_score(y_test, y_pred_lr))
print("F1:", f1_score(y_test, y_pred_lr))
print("ROC-AUC:", roc_auc_score(y_test, y_prob_lr))
```

7. Random Forest

None

```
rf = RandomForestClassifier(  
    n_estimators=100,  
    random_state=42  
)  
  
rf.fit(X_train,y_train)  
  
y_pred_rf = rf.predict(X_test)  
y_prob_rf = rf.predict_proba(X_test)[: ,1]
```

Metrics:

None

```
print("Accuracy:", accuracy_score(y_test,y_pred_rf))  
print("Precision:", precision_score(y_test,y_pred_rf))  
print("Recall:", recall_score(y_test,y_pred_rf))  
print("F1:", f1_score(y_test,y_pred_rf))  
print("ROC-AUC:", roc_auc_score(y_test,y_prob_rf))
```

8. Cross Validation

None

```
cv_lr = cross_val_score(  
    lr,  
    X,  
    y,  
    cv=5,  
    scoring='accuracy'  
)  
  
cv_rf = cross_val_score(  
    rf,  
    X,  
    y,  
    cv=5,
```

```
        scoring='accuracy'
    )

    print("LR CV Accuracy:", cv_lr.mean())
    print("RF CV Accuracy:", cv_rf.mean())
```

9. Confusion Matrix Plot

```
None
import matplotlib.pyplot as plt
from sklearn.metrics import ConfusionMatrixDisplay

ConfusionMatrixDisplay.from_estimator(
    rf,
    X_test,
    y_test
)

plt.show()
```

10. ROC Curve

```
None
from sklearn.metrics import roc_curve

fpr, tpr, _ = roc_curve(y_test, y_prob_rf)

plt.plot(fpr, tpr)
plt.xlabel("False Positive Rate")
plt.ylabel("True Positive Rate")
plt.title("ROC Curve")
plt.show()
```

Short Report Structure

Introduction

The goal of this project is to predict customer churn using machine learning classification techniques.

Data Preprocessing

- Removed/handled missing values.
- Encoded categorical variables.
- Split data into training and testing sets.

Models Used

- Logistic Regression
- Random Forest

Evaluation Metrics

- Accuracy
- Precision
- Recall
- F1 Score
- ROC-AUC

Results

Model	Accuracy	Precision	Recall	F1	ROC-AUC
Logistic Regression	xx	xx	xx	xx	xx
Random Forest	xx	xx	xx	xx	xx

Conclusion

Random Forest achieved the best overall performance and was selected as the final model due to higher ROC-AUC and F1 score.